CSCI 4551:
Group 12

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Motivation:

Problem

- This ROS2 system solves the problem of an active search in an unknown environment. The robot has to build its map of the world while finding it's target(s), and it has to be able to seamlessly switch between different targets that may or may not have been found already.
- We abstracted our targets and the target selection mechanisms to being color-based, using flashcards to drive our inputs to the system. In a more utility-focused system, this could be search and rescue operations, agricultural operations (mapping orchards, identifying fruit that can be harvested, picking the fruit), perhaps doing inspections or cleaning, etc.

Goals

- Use computer vision
- Use SLAM
- Avoid gestures

Concerns

- Using colored flash cards may be affected by background noise
- It may also require printing in color or using random, non-uniform objects

Relevance to the Course:

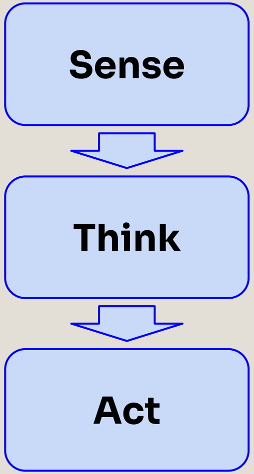
ROS2 System Integration
- Modular, node-based software design
- Publish / Subscriber nodes

Autonomous Navigation with Nav2
- Utilized for path planning, obstacle avoidance, and goal execution

Explore_Lite
- Drives Nav2 to explore unmapped “frontiers”

SLAM
- Build a map and track robot position in real time

Computer Vision
- Color-based object detection
- OpenCV and Cv_Bridge



Project Outline / Design:

Phase 1 – Set up robot environment
- Turtlebot3 house
- Added colored blocks around the house

Phase 2 – Robot platform setup
- Turtlebot3 burger robot
- Added a camera for color detection
- Integrated robot with ROS topics, nodes, and sensor data
- Initial waypoint-based navigation system

Phase 3 – Robot navigation
- Used Nav2 to move the robot through the house
- Enabled path planning
- Allowed the robot to autonomously explore
- Use explore_lite node to implement frontier-exploration

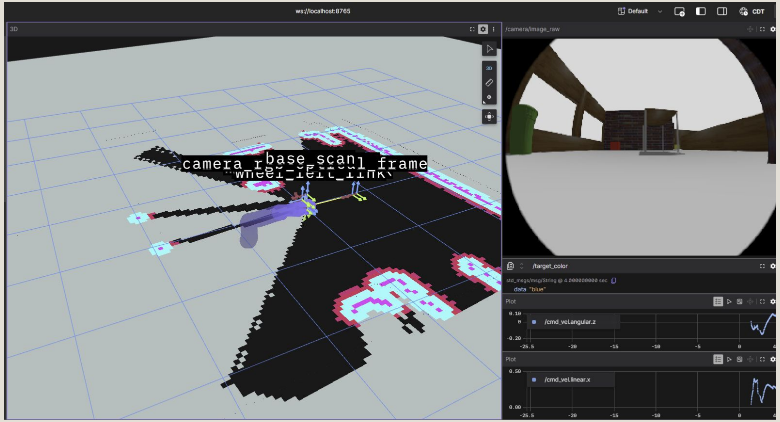
Phase 4 – Color Detection
- Used camera data with CV bridge and OpenCV
- Processed images to detect specific colors

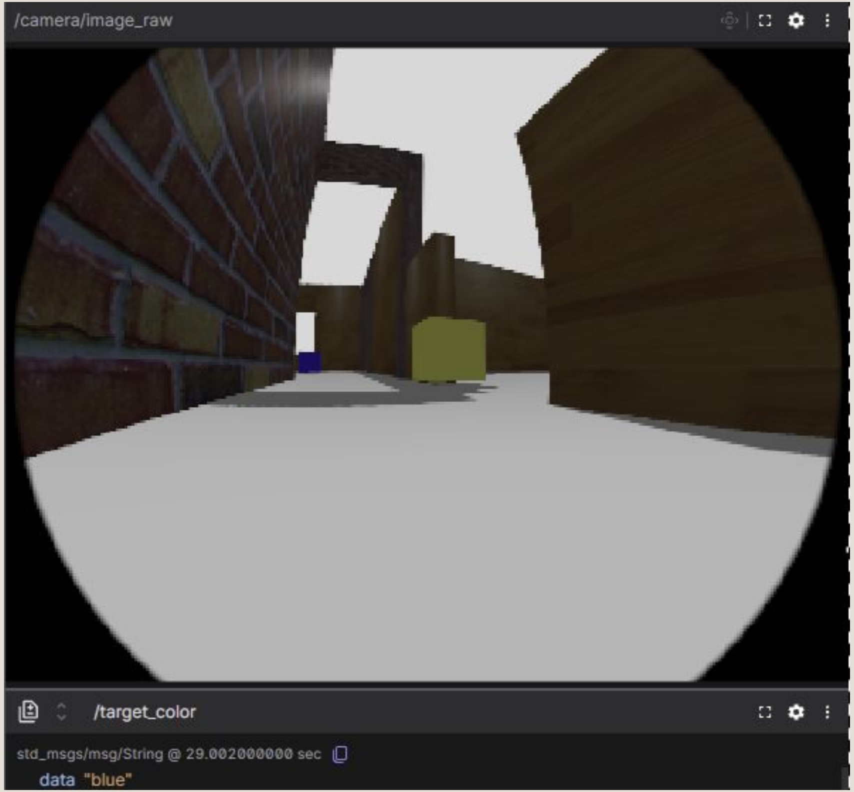
Phase 5 – Cube Location Memory
- Stored detected cube locations passed on to robot’s pose
- Robot can go back to those locations when assigned a new color
- Outputs location data for visualizations

Phase 6 – Continue development and testing
- Video feed to scan flashcards
- Try robot out on a different environment
- Use different objects besides cubes

Implementation - Gazebo Environment:

- **Exploration Environment:** turtlebot3_house.world
- Burgerbot spawns in at (x = -6.0, y = 3.0, z = 0.01)
- Four colored cubes are spawned throughout the Map:
  - Red: (x = -1.5, y = 4.0)
  - Yellow: (x = 1.6, y = -0.2)
  - Blue: (x = 4.7, y = 1.0)
  - Green: (x = 6.0, y = -1.2)
- The colored cubes are spawned at z = 0.2

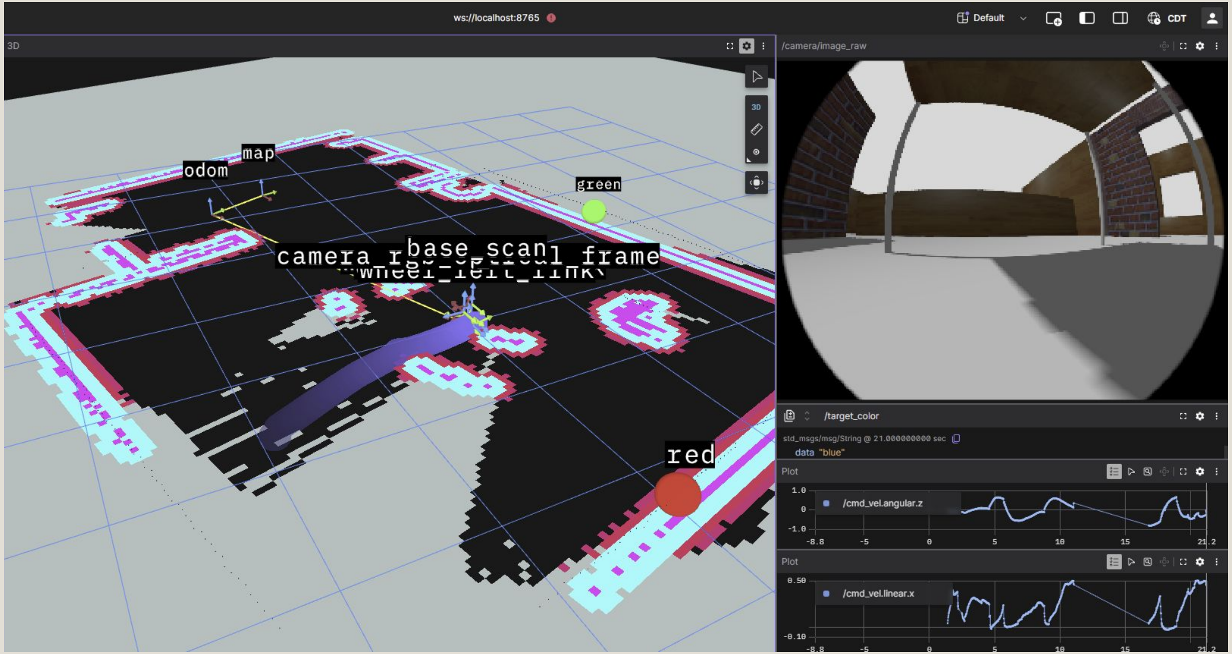




Nav2 and Explore_Lite

The basic cycle for automatically exploring the map:

1) explore_lite first picks a frontier (edge of known free space in the cost map)
2) Nav2 drives the robot there.
3) While driving and after arriving, the robot's lidar scans new areas.
4) SLAM integrates those scans, growing the costmap.
5) New frontiers appear, gradually bringing us deeper into the formerly-unknown space.
6) explore_lite picks one of those frontiers, and we go back to step 2.



Work Distributions:

- **Nick Empson:** Phases 1 and 2 - Set up the environment, adding the spawn information, adding the camera for in-simulation color detection, integrating the robot with the ROS system, implementing the initial waypoint system.

- **Tyler Hamilton:** Phases 3 and 5 - Using Nav2 to move the robot through the house, explore_lite to drive Nav2 using costmap data, storing detected colored cube locations, and alternating between exploring the map when we haven't discovered the location of the current goal yet or going to a stored memory location if we have.

- **Jason Guoin:** Phases 4 and 6 - Processing images using camera stills to detect specific colors, color detection using a video feed to actively detect colors from flashcards.

Current Topic Map, Future Additions:

Mandatory

- Detecting words on flash cards in camera feed
- Using those flashcards to select the target color
- Finish documentation

Stretch Goals

- Create new environments
- Add more varied objects